

Device Description: Heel Cuff Wearable Motion Sensor

1. Device Overview

The Heel Cuff Wearable Motion Sensor (Heel Cuff Device) is a small, lightweight, non-invasive wearable device designed to collect lower-limb motion data during pickleball play and can be extended to sport-related movement and daily activities such as walking. The device is worn externally on the participant's heel and is used solely for research data collection purposes.

The device does not provide feedback, stimulation, diagnosis, or treatment. It functions similarly to a consumer fitness tracker but is positioned on the heel to enable more precise measurement of ankle motion, particularly ankle flexion and extension during movement.

2. Purpose of the Device

The purpose of the Heel Cuff device is to record ankle motion data, specifically ankle flexion angle, during walking and pickleball-related movement tasks. These data are used to analyze ankle mechanics and to compare observed motion patterns to ranges reported in biomechanics literature.

The device is used for research and exploratory analysis only. It is not intended to diagnose injuries, assess clinical conditions, or provide medical recommendations to participants.

3. Physical Description

The device consists of:

- A flexible outer housing that conforms to the heel
- An internal inertial motion sensor used to measure orientation and movement
- A small microcontroller for data collection and storage
- A low-voltage rechargeable battery

All electronic components are fully enclosed within the housing. The device has no exposed wiring, sharp edges, or moving parts. Components that contact the participant are made from flexible polymer materials designed to prioritize comfort and reduce irritation during movement.

A mechanical assembly drawing of the device is provided in Appendix A to illustrate its construction and component layout.

4. How the Device Is Worn and Used

1. The device is placed on the participant's heel, typically over a sock or shoe.
2. Flexible arms hold the device in place without straps or adhesives.
3. The participant performs walking or pickleball-related movement tasks while motion data are collected.
4. The device remains external and is never implanted or inserted.
5. After the trial, the device is removed immediately.

A researcher is present during all use of the device.

5. Data Collection and Handling

The device collects motion and orientation data from onboard sensors to calculate ankle flexion angle during movement. Data is temporarily stored on the device and later transferred to a research computer for analysis.

- No audio, video, or GPS data is collected.
- No personally identifiable information is recorded by the device.
- The device does not transmit data to external servers or cloud services.

Data is analyzed collectively to examine movement patterns and trends across participants.

6. Safety and Risk Assessment

The Heel Cuff device is considered minimal risk.

Electrical Safety

- Powered by a low-voltage (3.7 V) rechargeable battery
- Fully enclosed electronics
- No electrical contact with the participant

Mechanical Safety

- Rounded edges
- Flexible materials in contact with the body
- No pinch points or sharp features

Thermal Safety

- The device does not generate noticeable heat during operation

Skin Contact

- The device is worn over clothing (sock or shoe)
- No direct prolonged skin contact required
- Constructed from non-latex materials

Failure Mode

If the device malfunctions, it will simply stop recording data and can be removed immediately without risk to the participant.

The risks associated with wearing the device are no greater than those associated with wearing a small consumer wearable or fitness tracker.

7. Duration of Use

Typical device use lasts between 5 and 30 minutes per participant. The device is removed immediately after data collection.

Participants may request removal of the device at any time without penalty.

8. Cleaning and Hygiene

Between participants, the device is:

- Wiped with disinfectant or isopropyl alcohol wipes
- Visually inspected for damage or wear
- Allowed to fully dry before reuse

9. Participant Exclusion Criteria (Device-Related)

Participants will not use the device if they have:

- Open wounds or skin irritation on the heel
- Known sensitivities to wearable materials
- Any condition that makes wearing the device uncomfortable or unsafe

10. Appendix

Appendix A: Mechanical Assembly Drawing of the Heel Cuff Wearable Motion Sensor

